



Student Performance Data Analysis Project

Project Overview

This project investigates how various lifestyle and background factors affect student academic performance. Using a dataset of 5,000 students, the goal was to discover meaningful patterns and correlations using **Google Sheets**, showcasing skills in **data cleaning, analysis, visualization, and formulas**.



Data Cleaning

A clean dataset was essential for accurate analysis. Steps included:

-  Verified no missing values using COUNTBLANK
 -  Checked and removed duplicate IDs with COUNTIF
 -  Corrected inconsistent spellings using filters (e.g., "CS" → "Computer Science")
 -  Renamed ambiguous values ("None" → "Unknown" in Parent Education)
 -  Removed unnecessary columns (email, ID, separate test scores)
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Exploratory Data Analysis (EDA)

Data Enrichment

- Categorized **Total Score** into: Low, Medium, High
- Categorized **Sleep Hours** into: Low, Moderate, Good, High
- Used IFS, COUNTIF, and COUNTIFS functions for categorization
- Created summary tables using **Pivot Tables**



Visualizations

Used various chart types to reveal trends:

- Pie Charts (e.g., department distribution)
- Bar and Stacked Bar Charts (e.g., stress vs performance)
- Summary tables with calculated % High Scores

Key Insights

1. **Sleep Quality Matters:** Students with good sleep had **higher chances** of scoring well.
 2. **Stress Has a Complex Impact:** Low and very high stress levels showed **more High performers**.
 3. **Parental Education:** Students with parents holding **Master's/PhD degrees** performed slightly better, though differences were minimal.
 4. **Extracurricular Activities:** Slightly **higher performance in students not involved**, suggesting involvement may not directly boost academic results.
 5. **Family Income:** Had **minimal impact** on academic outcomes – all income groups performed similarly.
 6. **Attendance:** No strong performance difference based on attendance level.
 7. **Department-wise Study Hours:** Slight variations in average weekly study hours, but not strongly tied to performance.
 8. **Medium Scores Dominate:** Across nearly all factors, the **majority of students fall into the Medium score category**.
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Summary Table Snapshot

Factor	Category	Total	% High Scores
Sleep Quality	Good	2072	37.64%
Stress Level	Level 1	521	44.53%
Parent Education	Master's Degree	1000	41.40%
Extracurricular	No	2488	41.56%
Family Income	High Income	1639	40.63%

 *This table gives a concise overview of how specific factors correlate with high academic performance.*

Tools & Skills Demonstrated

- **Google Sheets:** Data entry, formula creation, advanced filtering

- **Functions Used:** COUNTIF, COUNTIFS, IFS, AVERAGEIF
 - **Pivot Tables:** To explore grouped relationships and summarize insights
 - **Charts:** Pie, bar, and stacked bar charts to enhance data storytelling
 - **Data Cleaning:** Handling duplicates, inconsistencies, irrelevant data
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Conclusion

This project highlights how multiple factors—sleep, stress, parental background, and activities—impact student success. While some influences like sleep and stress levels showed clear trends, others like income or parental education had limited effects.